

Satellite Challenge Standards Manual (SCSM)

Space Challenge
Edition 7 | Revision 2



The electronic version is the official, approved document.
Ensure you are using the correct version before proceeding.

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Record of Revisions

The Satellite Challenge Standards Manual (SCSM), previously known as the LASC Rules & Requirements Document, has been revised to streamline the documentation system by consolidating information into a single document.

As a result, the rules and requirements for mission organization, as well as the technical specifications for the satellite projects, are now available in this document. The following table records all revisions of the document since the first Latin American Space Challenge.

Edition Number	Revision Number	Issue Date	Effective Date
1	N/A	N/A	N/A
2	2	February 2020	February 2020
3	1	July 2021	July 2021
4	1	January 2022	January 2022
5	2	January 2023	May 2023
6	1	June 2024	June 2024
6	2	May 2025	May 2025
6	3	June 2025	June 2025
7	1	March 2026	March 2026
7	2	June 2026	June 2026

SCSM Seventh Edition

The following table describes changes contained in Edition 7 of the Satellite Challenge Standards Manual (SCSM Ed. 7). The table describes significant changes and individual changes in each of the SCSM sections and provisions.

SCSM 7 Revision 2 Highlights	
Area(s) of Change	Description(s)
Section 6.1. CanSat Mass, Volume, and Interface Requirements.	The minimum and maximum CanSat mass requirements have been updated to streamline the design specifications.

Introduction

1. Purpose

The Latin American Space Challenge (LASC) Satellite Challenge Standards Manual (SCSM) is published to establish the standards, guidelines, and operational protocols necessary for participants to effectively engage in the LASC Satellite Challenge.

This manual serves as the definitive reference for teams preparing for and participating in the competition, outlining the criteria and practices to be followed during the design, development, assembly, integration, and potential launch & recovery of the satellite project. It also provides a structured approach to align with best practices in space engineering and mission planning.

2. Structure

The LCSM is based on the LASC Documentation System described in Section 2 of the LASC General Rules (LGR) document. This documentation system is structured into three hierarchical levels of documents that establish the overall framework of the event and its challenges.

Adherence to this documentation system by teams and participants is mandatory in order to ensure safety, organization, and compliance with event requirements.

The LASC SCSM is organized as follows:

- Section 1 → Satellite Challenge (SAT);
- Section 2 → Mission Organization (ORG);
- Section 3 → Safety (SAF);
- Section 4 → General Requirements (GEN);
- Section 5 → CubeSat Requirements (CUB);
- Section 6 → CanSat Requirements (CAN);
- Section 7 → PocketQube Requirements (PQB);
- Section 8 → Satellite Structure and Materials (SSM);
- Section 9 → Electronic Power System and Communications (ECS);
- Section 10 → Recovery System (REC).

Each section in this Manual is assigned an associated 3-letter identifier (in parentheses above). The reference number for every standard or recommended practice within a section includes the specific 3-letter identifier for that section (e.g., SAT 1.1.1).

3. LASC Standards and Recommended Practices (LSARPs)

LSARPs in this manual are designed for use within the Latin American Space Challenge (LASC) and provide the criteria for evaluations. The LASC General Rules (LGR) document defines the Documentation System and LSARPs.

4. Manual Revisions

The SCSM is subject to an annual revision, which invariably leads to a new edition of the SCSM. In the event that critical issues emerge impacting the content of the SCSM, a revision to the current edition will be undertaken.. All changes in this document are listed in the revision highlights table. For easier orientation, the following symbols identify any changes made within each section:

- △ Addition of a new item.
- ↻ Change to an item.
- ⊗ Deletion of an item.

5. Acronyms and Definitions

AGL - Above Ground Level: The altitude of the rocket measured vertically from the launch pad ground elevation to the vehicle position or apogee.

All altitude requirements in the Rocket Challenge shall be measured in meters Above Ground Level (AGL) from the launch pad elevation. GPS receivers typically measure altitude relative to Mean Sea Level (MSL), therefore teams shall ensure proper conversion when reporting altitude data.

Caliber: A unit of length equal to the external diameter of the airframe body tube.

CG - Center of Gravity: The point at which the total mass of the vehicle is considered to be concentrated. The vehicle will balance at this point in any orientation.

CP - Center of Pressure: The point along the vehicle where the resultant aerodynamic forces are considered to act. The relative position between the Center of Pressure and the Center of Gravity determines the aerodynamic stability of the vehicle.

EGM - Event Guidance and Milestones: LASC Event Guidance and Milestones document available in the Documentation section of the LASC website.

LCC - Launch Control Center: The designated area from which launch operations are monitored and controlled, including vehicle status monitoring, safety coordination, and launch authorization.

LGR - LASC General Rules: LASC General Rules document available in the Documentation section of the LASC website.

LRR - Launch Readiness Review: A formal review conducted to verify that the vehicle, team, and operations are ready for launch.

LSARP - LASC Standards and Recommended Practices: Standards and recommended practices defined and detailed in the LASC General Rules (LGR) document.

MCC - Mission Control Center: The designated area from which mission operations are monitored and controlled after launch, including telemetry monitoring, flight tracking, command operations, and mission status management.

MSL - Mean Sea Level: Altitude measured relative to the average sea level, commonly used in aviation and GPS altitude measurements.

RAA - Rocket Assembly Area: Designated area where teams assemble, integrate, and prepare their rockets for flight.

RBF - Remove Before Flight: Safety tag or device that must be removed prior to launch or flight operations.

SF - Safety Factor: The ratio between the maximum load or stress a component can withstand and the maximum load or stress expected during operation. The safety factor is used to provide a margin against uncertainties, material variability, and unexpected loads.

Selected Mission: A mission proposal submitted by a team that has been evaluated, approved, and officially accepted by the LASC Organization for participation in the event.

SMS - Safety Management System: A systematic approach to managing safety, including organizational structure, responsibilities, policies, procedures, risk management, and continuous safety improvement processes.

SRAD — Student Researched and Developed: A designation for hardware, software, or systems that have been primarily designed, developed, and built by students, rather than purchased as complete commercial off-the-shelf solutions.

Static Stability Margin: The distance between the Center of Gravity and the Center of Pressure divided by the airframe diameter, expressed in calibers.

Section 1 — Satellite Challenge (SAT)

Section 1 defines the Satellite Challenge (SAT) mission categories, administrative requirements for application and registration, covers each technical deliverable, and details all awards and team eligibility.

1.1. Challenge Overview

The Satellite Challenge is a competition for teams to design a nano or pico satellite mission. The primary objective of each mission is to progress from the initial concept stage to a potential launch on-board of an experimental rocket.

The challenge will recognize and reward teams that successfully pass the design review gates increasing their Technology and Operations Readiness Level, with the potential for their satellite to be launched by one of the experimental rockets of the Rocket Challenge.

Mission Categories & Objectives

SAT 1.1.1. Teams competing in the Satellite Challenge shall design a satellite project in one of the following mission categories:

- PocketQube or CanSat satellite project; or
- CubeSat satellite project.

The Satellite Challenge will have a minimum mission objective for each mission category. This objective can be the sole mission or part of a larger and more complex mission. The objectives for each mission are established below:

SAT 1.1.2. The PocketQube or CanSat Mission shall at least collect air temperature, humidity and pressure. The data may be stored on-board or transmitted by radio to a ground station.

SAT 1.1.3. The CubeSat Mission shall at least collect air temperature, humidity, pressure, inertial, and GPS data. The data may be stored on-board or transmitted by radio, and is analyzed after landing.

LASC will accept registrations for the Satellite Challenge of teams competing only with a satellite project (i.e. an experimental rocket project is not mandatory for this challenge), but teams are encouraged to participate in both challenges or partner with other team for a complete space mission (i.e. rocket launch carrying a satellite).

Teams that have missions selected for both the Satellite and Rocket Challenge are encouraged to launch the experimental rocket with the satellite project onboard. Bonus points may be awarded for successfully integrating missions from both challenges into a single launch, regardless of whether they are from the same team or different teams.

Onsite Event Participation

Teams with selected Satellite Missions are not required to participate in the onsite event. However, onsite participation fosters an environment of knowledge sharing and provides an opportunity for teams to present their work.

The Satellite Challenge will evaluate each selected mission through the design review stages and **maturity level gates** as explained in the Technology and Operations Readiness Level subsection of the [Appendix B: Detailing Grading Criteria](#). The further a mission progresses through these gates, the more points it can receive. The design review stages and maturity level gates are detailed in the following sections.

SAT 1.1.4. Teams with at least one selected satellite mission should participate in the onsite event to foster an environment of knowledge sharing and to potentially present their work.

SAT 1.1.5. Teams participating in the Satellite Challenge should prospect an experimental launch with teams participating in the Rocket Challenge.

1.2. Milestones

The Latin American Space Challenge features a timeline that categorizes milestones as either remote or onsite. Remote milestones refer to pre-event activities that take place before the onsite portion of the event. These include application and registration, progress updates, marketing actions, podium session materials, and mission report submissions. The master schedule for all milestones is detailed in the Event Guidance and Milestones (EGM) document.

Each remote milestone is outlined in this section, detailing the requirements for each submission and action leading up to the onsite event. [Section 3](#) provides safety provisions that shall be followed throughout the mission development process.

SAT 1.2.1. All Selected Missions shall meet a minimum quality standard for all milestones to secure participation in the onsite event, including the assignment of a launch slot.

1.3. Challenge Deliverables

The following sections define the deliverable materials event officials require from teams competing in the Satellite Challenge, including each deliverable's format and content.

All deliverables will be submitted to LASC per the instructions provided to the teams. Each relevant deliverable description will facilitate submission of that deliverable or will be communicated to teams as is determined by LASC Organization.

The scheduled due dates of all required deliverables are recorded in the EGM document, maintained on the LASC website.

Progress Updates

There will be four Progress Updates submitted per Selected Mission due prior to the event to track mission progress in design and development. These forms will contain questions regarding safety, organization, operations and testing, structures, power systems, data handling, sensors, communications, and recovery. The template for each progress update will be available at least two weeks before the deadline to allow teams sufficient time to collect the appropriate information and data.

SAT 1.3.1. Teams shall submit Progress Updates for each Selected Mission via the Latin American Space Challenge website (<https://www.lasc.space/>) on 4 (four) specific occasions prior to the onsite event: the Progress Update 1, Progress Update 2, Progress Update 3 and the Progress Update 4.

These Progress Updates will record progression in the project's technical characteristics during development. Event officials understand not all technical details will be known until later in the design process. Therefore, the Progress Updates prior to the final submission will be evaluated based only on their timeliness and completeness.

SAT 1.3.2. Total completeness of the Progress Update form is required at all times. Reasonable engineering estimates and approximations are expected during the application process, but will be subject to progressive additional scrutiny in the subsequent Progress Updates.

Teams should briefly mention their ongoing discussions and analysis in the comment fields for any numerical submissions that are known to be unreasonable or remain undecided.

Teams may also respond to undecided criteria by demonstrating their understanding of any applicable event guidance or best practice governing the particular detail.

SAT 1.3.3. Teams should briefly mention their ongoing discussions and analysis in the comment fields for any numerical submissions that are known to be unreasonable or remain undecided.

Teams may also respond to undecided criteria by demonstrating their understanding of any applicable event guidance or best practice governing the particular detail.

Marketing Actions

Since the early days of the Apollo missions, the general public has been captivated by the spectacle of rocket launches, eager for more information and the chance to witness these events firsthand. The allure of a launch vehicle is immense, drawing thousands of people to travel far from home to be near launch sites, just to hear the thunderous roar of rocket engines.

Event officials recognize the importance of structured marketing efforts to promote each team and its members. These efforts aim to increase visibility, attract attention to the missions, and help support fundraising for each project. Such efforts are essential to realizing the dream of having thousands of spectators watch experimental rockets and satellites launch from LASC.

The Marketing Actions are divided in three deliverables: a Mission Patch, a Pitch Video and a competition for likes/followers in the following social media: Instagram, TikTok, X (Twitter), Facebook and LinkedIn.

Mission Patch

Mission patches are emblems designed and worn by astronauts and people affiliated with a mission, such as Rocketeers and Satelliteers. The patches depict an image associated with the mission.

SAT 1.3.4. Each team shall submit an appropriate Mission Patch for each Selected Mission during the Progress Update 1. Submissions shall be made in a timely manner using the designated form on the LASC Website, following the schedule outlined in the deliveries roadmap.

A Mission Patch must contain the mission and event references. The Mission Patch must include, at a minimum:

- The Mission ID and mission reference;
- The event reference; and
- The year of the event.

For more information and examples, teams should access the following websites:

- Wikipedia - Mission Patch: https://en.wikipedia.org/wiki/Mission_patch
- Human Spaceflight Mission Patches: <https://www.nasa.gov/gallery/human-spaceflight-mission-patches/>
- NASA - Houston We Have a Podcast, Mission Patches: <https://www.nasa.gov/podcasts/houston-we-have-a-podcast/mission-patches/>
- National Air and Space Museum - The Meaning of Mission Patches: <https://airandspace.si.edu/stories/editorial/meaning-mission-patches>
- SpaceX Unofficial Patch List: <https://spacexpatchlist.space/>
- Blue Origin's NS-30 Mission Patch: <https://www.instagram.com/p/DGb0jnkNspS/>

The team's logo will not be accepted as a mission patch. Late submission will be accepted, but the mission may be penalized by not timely submitting a Mission Patch.

Pitch Video

Reels, short videos, and pitch presentations—these are new terms that have become part of our daily lives. The younger generations are completely immersed in social media, especially videos: easy to watch and fun! To make the competition more diverse and not just about technical development, it is crucial to communicate what each selected mission is doing to the broader society. Therefore, each selected mission will be required to record a pitch video explaining their goals, concept of operations, and anything else the team feels needs to be communicated.

SAT 1.3.5. Each selected mission shall submit a Pitch Video explaining the mission, goals, concept of operations, each subsystem of the project and other important information and data.

SAT 1.3.6. The Pitch Video shall have a minimum duration of 2 minutes and a maximum duration of 5 minutes.

SAT 1.3.7. The Pitch Video shall have clear audio and video quality sufficient for proper evaluation.

SAT 1.3.8. The Pitch Video should include images, videos, animations, or footage of the team, the project, hardware, simulations, or development activities.

SAT 1.3.9. On or before a specified date prior to the event, teams shall submit the YouTube link of the video using the appropriate location indicated on LASC website.

The video shall be uploaded on YouTube with a title as "Mission Your_Mission_ID Pitch Video to the Event Edition LASC". For example, a team assigned the Mission ID "19", competing in the 7th LASC, would subtitle their YouTube Video as "Mission 19 Pitch Video to the 7th LASC".

Lights, Camera, Action!

Don't forget, this is a healthy competition! In order to grow the audience and get more attention to the space activities in the region, teams will be asked to conduct a marketing campaign to get likes and followers in the main social media. On a certain date, LASC will collect a frame of all teams' social media total followers. Then, from a specified time window, teams will be required to post and organically promote their selected missions for the Latin American Space Challenge.

SAT 1.3.10. Each team should promote the selected mission by posting reels, stories, shorts, videos, tweets, and other types of publicly accessible content on YouTube, Facebook, Instagram, LinkedIn, TikTok, and/or X (Twitter). The appropriate content shall be posted during the time windows specified in the LASC Event Guidance and Milestones (EGM).

SAT 1.3.11. Each post must include the Official LASC Event Patch in the image or video, mention the LASC website in the description, and tag the official LASC profile. Each post can be in any official language of the event.

SAT 1.3.12. Each mission leader should timely submit link(s) to the post(s) related to the *Lights, Camera, Action!* to be analyzed by event officials by the appropriate form available on LASC website. There will be a limit of 20 links for each selected mission.

Event officials will collect the total number of followers for each team immediately before and after the milestone window, as well as the total number of likes, comments and other public insights on posts about the selected mission, to score the "Lights, Camera, Action!" deliverable. The score will be normalized from the team with the greatest marketing reach to the team with the least for each selected mission.

Mission Report (MR)

Each team shall submit a Mission Report describing their mission objectives, the experimental satellite project and each module or system, and the concept of operations (ConOps) to the technical evaluation board and event officials. This document shall include detailed information on the design review, goals, tests, and the most accurate technical data of the satellite.

SAT 1.3.13. Teams shall timely submit a digital, PDF copy of their Mission Report for each Selected Mission, with the file name "Your Mission ID_Mission Report". For example, a team assigned the Mission ID "19" would submit a digital copy of their Mission Report using the filename "19_Mission Report".

SAT 1.3.14. The Mission Report shall be formatted according to the style guide of the Latin American Space Challenge (LASC), using a provided Microsoft® Word document template. Always check the template maintained on the website before drafting your Mission Report to ensure you are using the latest version.

SAT 1.3.15. The Mission Report shall be no longer than 50 pages, including figures, footnotes, sources, source endnotes, nomenclature lists, equations, explanations of variables etc. This does include the Appendices. However, appendices are not necessarily read in detail by the event officials.

SAT 1.3.16. The Mission Report's main title is left to the team's discretion, however; the paper shall be subtitled "Mission Report Your Mission ID to the Event Edition Latin American Space Challenge".

For example, a team assigned the Mission ID "19", competing in the 7th LASC, would subtitle their Mission Report "Mission Report 19 to the 7th Latin American Space Challenge".

Further requirements are given in [Appendix A: Details for the Mission Report](#), including the required minimum Mission Report sections and appendices. Additional sections, subsections, and appendices may be added as needed.

Presentation Session

Event officials aim to promote networking and knowledge sharing between teams during the onsite event. Podium Sessions will be held in designated slots to foster an environment of technology development. Teams with selected missions may register to participate as speakers in a Podium Session, discussing specific developments of their mission.

Selected missions chosen to speak in the Podium Session may receive additional points in the evaluation of their design implementation. The date of the official announcement of selected missions chosen to speak in the Podium Session will be published in the LASC Event Guidance and Milestones (EGM) document.

SAT 1.3.17. Each mission willing to participate in the Podium Session should submit a Podium Session Abstract on a particular aspect of their work for competition officials and the judging panel to consider including in a Podium Session held during the on-site event.

The file name should be "Your Mission ID_Podium Session Slides." For example, a team assigned the Mission ID "19" would submit their slides using the filename "19_Podium Session Abstract".

SAT 1.3.18. Each mission willing to participate in the Podium Session should submit a digital PDF copy of their Podium Session Slides for a specific Selected Mission to the appropriate LASC submission form available on the website.

The file name should be "Your Mission ID_Podium Session Slides." For example, a team assigned the Mission ID "19" would submit their slides using the filename "19_Podium Session Slides."

SAT 1.3.19. The Podium Session Abstract shall follow the formatting guidelines provided in the LASC Podium Abstract Template, available on the LASC website. There is no template for the Podium Session Slides in order to preserve each team's creativity.

However, all slides must include the Official LASC Event Patch in one of the corners to clearly identify the event in the podium materials.

SAT 1.3.20. The Podium Material's main title is left to the team's discretion. However the first slide shall be subtitled "Podium Session Your Mission ID to the Edition LASC". For example, a team assigned the Mission ID "19", competing in the 7th LASC, would subtitle their Podium Slide "Podium Session for Mission 19 to the 7th LASC".

SAT 1.3.21. The Podium Material can be given in English, Portuguese, or Spanish; however, the slides shall be in English. Teams should inform the language to be used during the Presentation Session in the appropriate LASC submission form. The language shall not be changed after submission.

SAT 1.3.22. Each Podium Session shall last between 7 and 10 minutes for the presentation, followed by up to 5 minutes for questions and answers, totaling a maximum session duration of 15 minutes.

LASC Officials will review all submissions and select the missions invited to present during the Podium Session. The selected missions will be announced in a livestream at least one week before the on-site event.

1.5. Scores, Evaluation and Awards

Event officials will evaluate competitors for Place Awards within each mission category based on the quality of required project documentation, the quality of their system's overall design and simulation, the mission's overall excellence, efficiency and performance demonstrated at the Mission Report, and finally, the launch operations and safety.

Scoring Categories and Grading Criteria

Missions will be scored in five different scoring categories or areas, which are (1) Team Effort, (2) the Mission Report, (3) the Design Implementation, (4) the Technology and Operations Readiness Level, and (5) Bonus Points. These are weighted according to the following table.

In each scoring category, a set of grading criteria is established. These criteria will be evaluated by the jury for each mission individually. Each grading criterion has several, more detailed, topics that establish what the jury will look for during the grading process.

Table 1: Weight of the Scoring Categories for the Satellite Challenge.

Scoring Categories	Possible Points	% of Total Points
Team Effort	100	10%
Mission Report	200	20%
Design Implementation	100	10%
Technology and Operations Readiness Level	550	55%
Bonus Points	50	5%
TOTAL	1000	100%

These detailed topics are weighed equally within each criterion, while the main criteria are weighted differently within each mission category. The details of the grading criteria can be found in [Appendix B: Detailed Grading Criteria](#).

1.6. Specific Awards

The Satellite Challenge awards First and Second Place in each category, as well as the Best Experimental Satellite of the Year. Overall Awards are defined in the LASC General Rules (LGR) document.

Best Experimental Satellite of the Year

One team among the First Place Award winners in each category defined in this document will be named the Best Experimental Satellite of the Year. The recipient of this award is determined by qualitative assessments of the event officials made throughout the entire event.

The Best Experimental Satellite of the Year not necessarily will be the highest scoring mission, but will be awarded to a single team chosen by the event officials based on project complexity, technology and innovation.

A team is considered eligible for the award after participating in the Satellite Challenge submitting all documents, reports, activities, and presenting the project.

Mission Category “Place” Awards

A First Place Award will be granted to the highest scoring, eligible team in each of the Mission Categories defined in this document. A Second Place Award will be granted to the 2nd highest scoring, eligible team in each Mission Category. A team is considered eligible for the place award(s) in its category after submitting all documents, reports and activities.

In the event, if no teams meet this definition in a given category, event officials may issue Category Place Awards at their discretion based on multiple factors – including points accrued, participation and engagement, and overall performance.

Section 2 — Mission Organization (ORG)

Section 2 — Mission Organization (ORG) has been moved to the LASC General Rules (LGR). All guidance related to team and mission organization, including applicable LSARPs and application and registration, is now consolidated in that document.

Section 3 — Safety (SAF)

Section 3 details the Safety (SAF) policy and provisions for all missions, members, and teams to ensure event safety. This section covers the general guidelines for Satellite Missions.

3.1. Safety Policy

Safety is our Value number 1.

The Safety Management System (SMS) of the Latin American Space Challenge is composed of four pillars: Safety Policy, Safety Risk Management, Safety Assurance, and Safety Promotion.

All team members shall report any observed unsafe activities to event officials. The event will provide a specific channel for these reports, which will be anonymized before any potential action is taken by event officials.

The event will be guided by a Just Culture approach, where errors will be addressed through reeducation. However, any violation of rules will be treated as potential unruly behavior, with penalties or disqualification possible as outlined in Section 2.

During the Launch Readiness Review (LRR), event officials will use a checklist to approve an experimental rocket for loading with energetics and a rocket motor. Ensuring safety will be a key process for issuing any Flight Card during the event.

Event officials will conduct a safety risk management analysis before each launch to reduce the potential probabilities and/or severities of any operational risks.

Promotion is key to raising the bar among all participating teams. Event officials will conduct safety briefings to increase overall public awareness. Launching an experimental rocket is not trivial; the event has associated risks. All participants shall understand and accept these risks by signing a Waiver and Release of Liability to participate.

SAF 3.1.1. All participants shall follow the Latin American Space Challenge Safety Policy.

SAF 3.1.2. Participants in the satellite mission will not engage in activities after potential integration with the experimental rocket but must adhere to the Safety Policy at all times. Participants engaged in both satellite and rocket missions shall coordinate their activities with event officials.

3.3. Mandatory Safety Guidance

SAF 3.3.1. No energetics can be stored and/or transported in closed vessels or any other condition that may cause pressurization.

SAF 3.3.2. The personnel involved in any operation or integration shall wear Personal Protective Equipment (PPE). The minimum PPE required are the helmet and a signaling vest. Additional PPE provided by the teams are welcome.

SAF 3.3.3. Satellites shall not contain any energetic devices.

SAF 3.3.4. Satellites shall not contain any propulsion system.

SAF 3.3.5. Satellite missions seeking to independently recover shall consider activation of the recovery system by the experimental rocket's electronics system.

SAF 3.3.6. The satellite shall have an external and easily accessible manual hardware kill switch that isolates battery power from the rest of the satellite payload system.

SAF 3.3.7. No exposed point or area of the satellite shall carry a live voltage at any point during operation of the satellite payload.

Section 4 — General Requirements (GEN)

Section 4 defines the General Requirements (GEN) overall guidance and requirements for the development of satellite projects, including material selection and prohibitions. The following provisions address key points applicable to all satellite projects independently of mission category and form factor.

4.1. General Satellite Mission Requirements

GEN 4.1.1. The total budget of the final satellite project should not exceed USD 1,000. Ground Stations (GS) and any related non-flying item will not be considered in the budget. In the case of sponsorship, all sponsored items should be specified in the budget with the actual corresponding costs on the market.

GEN 4.1.2. The Satellite Mission shall comply with the minimum mission objectives stated on [Section 1](#).

GEN 4.1.3. The Satellite Mission shall ensure no deliberate detachment of any components throughout the lifetime of the entire mission, including launch, ejection (optionally) and recovery.

GEN 4.1.4. The Satellite Project should be able to withstand an acceleration of up to 20 g. This provision is complemented in [Section 6](#).

GEN 4.1.5. Inclusion of a positioning system for retrieval (beeper, radio beacon, GPS, etc.) is recommended.

4.2. Forbidden Materials

GEN 4.2.1. Explosives, detonators, pyrotechnics, and inflammable or dangerous materials are strictly forbidden. All materials used shall be safe for personnel, equipment, and the environment. In case of doubt by LASC, Material Safety Data Sheets (MSDS) may be requested from the teams.

GEN 4.2.2. Materials that can be toxic, flammable or potentially hazardous shall not be used. The use of Li-Ion batteries is exempted from this constraint, provided that there is adequate prevention against thermal runaway.

4.3. Electrical System and Frequency Guidelines

GEN 4.3.1. The Satellite Project shall be powered by a battery and/or solar panels. It should be possible for the systems to remain switched on for 4 (four) continuous hours.

GEN 4.3.2. The battery shall be easily accessible in case it has to be replaced/recharged.

GEN 4.3.3. The Satellite Project shall have an easily accessible master power switch.

GEN 4.3.4. The antennas, transducers and other elements of the Satellite Project shall not extend beyond the satellite's maximum diameter until it has left the experimental launch vehicle.

GEN 4.3.5. The assigned frequency should be respected by all missions in the Launch Campaign. It is recommended that teams pay attention to the design of the Satellite Project in terms of hardware integration and interconnection, so the radio frequency can be easily modified if necessary.

4.4. Launch Readiness Review (LRR)

GEN 4.4.1. All Satellite Missions willing to be evaluated and/or launched during the Latin American Space Challenge shall be prepared for a Launch Readiness Review (LRR).

GEN 4.4.2. Missions willing to launch their satellite projects during the Latin American Space Challenge shall request the initial Launch Readiness Review (LRR) together with the experimental rocket that will be the launch vehicle.

If a satellite mission is not ready or fails to present the project at the same time of the experimental rocket mission, the launch of the satellite will be denied.

GEN 4.4.3. Missions not willing to launch their satellite projects may request a Workbench Evaluation (i.e., Launch Readiness Review) at the Mission Control Center (MCC) at any time.

GEN 4.4.4. The Launch Readiness Review (LRR) will take place at the event officials' tent in the Rocket Assembly Area (RAA) designated as Mission Control Center (MCC). The LRR must be requested by the Mission Leader.

During the first request for an LRR, the mission will be placed in a queue with a determined priority. Subsequent requests will follow the queue based on the assigned slot, position in the process, and mission complexity.

GEN 4.4.5. Teams shall ensure that the satellite is in an LRR-ready state before the first evaluation or if it is a revisit after an "orange" LRR Status. The Launch Readiness Review (LRR) will follow a checklist based on the provisions of this document. Each item on the checklist will serve as a criterion for the review.

All criteria of the LRR can be scored as "red" (Denied), "orange" (Action Needed), "yellow" (Provisional), "green" (Approved), or "gray" (Not Applicable). The Launch Readiness Review (LRR) details can be found in the Chapter 4 of the LASC Rocket Challenge Standards Manual (RCSM).

GEN 4.4.6. If a Satellite Mission fails to pass the Launch Readiness Review (LRR), the experimental rocket mission will not be affected. Conversely, if the experimental rocket mission fails, the Satellite Mission will remain unaffected. However, scores may be deducted accordingly.

Section 5 — CubeSat Requirements (CUB)

Section 5 defines the specific requirements for CubeSat (CUB) satellite projects, including form-factor dimensions, mass, specific guidance, and other related criteria.

5.1. CubeSat Mass, Volume, and Interface Requirements

CubeSats are nanosatellite platforms based on a cubic-shaped form factor (approximately 100x100x100 mm per unit).

The external dimensions and typical maximum mass are slightly different depending on the number of units, as outlined in the following for the 1U, 2U and 3U cases.

CUB 5.1.1. The only CubeSat form-factors permitted for the Latin American Space Challenge are 1U, 2U, and 3U.

Table 2. CubeSat Dimensions.

Number of Units (U)	External Dimensions (mm)	Minimum Mass (g)	Maximum Mass (g)
1U	100x100x100	700	1330
2U	200x100x100	1400	2660
3U	300x100x100	2800	4000

CUB 5.1.2. Deployables such as antennas shall be constrained by the CubeSat, not the dispenser.

CUB 5.1.3. The CubeSat center of gravity should fall within the ranges specified in the following table:

Table 3: Ranges of Acceptable CG Locations as Measured from the Geometric Center on each Axis.

Number of Units (U)	X axis (cm)	Y axis (cm)	Z axis (cm)
1U	+ 2 / -2	+ 2 / -2	+ 2 / -2
2U	+ 2 / -2	+ 2 / -2	+ 4.5 / -4.5
3U	+ 2 / -2	+ 2 / -2	+ 7 / -7

CUB 5.1.4. The CubeSat structure should be made from aluminum alloy. Typically, Aluminum 7075, 6061, 6082, 5005, and/or 5052 are used for both the main CubeSat structure and the rails.

CUB 5.1.5. Rails should have a minimum width of 8.5 mm measured from the edge of the rail to the first protrusion on each face. Rails should have a surface roughness less than 1.6 μm . The edges of the rails should be rounded to a radius of at least 1 mm. The ends of the rails on the +/- Z face shall have a minimum surface area of 6.5 mm x 6.5 mm contact area with neighboring CubeSat rails.

Section 6 — CanSat Requirements (CAN)

Section 6 defines the specific requirements for CanSat (CAN) satellite projects, including form-factor dimensions, mass, specific guidance, and other related criteria.

6.1. CanSat Mass, Volume, and Interface Requirements

CanSats are picosatellite platforms based on a cylinder-shaped form factor. The challenge is to fit all the major subsystems, such as power, sensors and a communication system, into this minimal volume.

CAN 6.1.1. The only permitted CanSat form-factor for the Latin American Space Challenge is 1C, as described in Table 4. All missions with a CanSat project shall comply with this established form factor.

Table 4. CanSat Dimensions.

Form Factor	Diameter (mm)	Height (mm)	Minimum Mass (g)	Maximum Mass (g)
1C	66	105	200	400

CAN 6.1.2. The mass of the CanSat must be between a minimum of 200g and a maximum of 400g. CanSats that are lighter must take additional ballast with them to reach the 200g minimum mass lower-limit required.

CAN 6.1.3. Radio antennas and GPS antennas can be mounted externally on the top or bottom of the can, depending on the design, but not on the sides.

CAN 6.1.4. Any antennas, transducers and other elements of the CanSat cannot extend beyond the can's diameter until it has left the launch vehicle.

CAN 6.1.5. Recommended materials for the structure are: PLA, ABS, PP e PETG. Potential metallic materials such as aluminum alloy may also be used.

Section 7 — PocketQube Requirements (PQB)

Section 7 defines the specific requirements for PocketQube (PQB) satellite projects, including form-factor dimensions, mass, specific guidance, and other related criteria.

7.1. PocketQube Mass, Volume, and Interface Requirements

PocketQubes are picosatellite platforms based on a cubic-shaped form factor (approximately 50x50x50 mm per unit) with each side one-half smaller than a CubeSat (100x100x100 mm per unit).

PQB 7.1.1. The only PocketQube form-factors permitted for the Latin American Space Challenge are 1P, 2P, and 3P.

Table 5. PocketQube dimensions.

Number of Units (P)	External Dimensions without Backplate (mm)	Sliding Backplate Dimension (mm)
1P	50x50x50	58x64x1.6
2P	50x50x114	58x128x1.6
3P	50x50x178	58x192x1.6

PQB 7.1.2. PocketQube-projects shall comply with the established minimum and maximum required mass detailed in Table 6.

Table 6. PocketQube minimum and maximum mass.

Number of Units (P)	Minimum Mass (g)	Maximum Mass (g)
1P	200	250
2P	400	500
3P	600	750

PQB 7.1.3. The envelope around the PocketQube shall be no more than 7 mm for components.

PQB 7.1.4. For appendages and deployables, maximum 10 mm is allowed. All deployable components shall be constrained by the PocketQube and not by the deployer.

PQB 7.1.5. The center of mass of the PocketQube shall not exceed 1 cm from its geometric center in a stowed position.

PQB 7.1.6. Recommended materials for the baseplate are: FR4 and Aluminum (7075, 6061, 6065, 6082).

PQB 7.1.7. Recommended materials for the structure are: PLA, ABS, PP e PETG. Potential metallic materials such as aluminum alloy may also be used.

PQB 7.1.8. Potential metallic materials used for the PocketQube that are in contact with the deployer and standoffs should be hard anodized.

PQB 7.1.9. All PocketQubes shall use at least two kill switches to keep the satellite offline while in the deployer.

PQB 7.1.10. Kill switches shall be located only on the Z-axis. There are two different possible placement areas:

- a) In the lateral side of the satellite within 20 mm from the Z-faces and touching the deployment rails;
- b) Aligned with the sliding backplate in Z-face and in contact with the PocketQube below or the pusher plate.

PQB 7.1.11. The kill switches shall not obstruct ejection of the satellite from the deployer and/or experimental rocket.

Section 8 — Satellite Structure and Materials (SSM)

Section 6 defines the Satellite Structure and Materials (SSM) overall guidance and requirements for the development of satellite projects, including material selection and prohibitions.

The following provisions address some key points applicable to the satellite projects, but are not exhaustive of the conditions affecting each unique design. Each team is responsible for thoroughly understanding, analyzing, and mitigating their design's unique load set.

8.1. Overall Structural Integrity

STR 8.1.1. Satellites shall be constructed to withstand the operating stresses and retain structural integrity under the conditions encountered during handling transportation, and the potential launch operation.

8.2. Material Selection and Tests

STR 8.2.1. Satellites shall be constructed from lightweight materials, such as 3D-printed low-temperature polymers or ductile lightweight metals like aluminum. The choice of materials and assembly techniques must be appropriate for the specific requirements to ensure optimal performance and safety.

STR 8.2.2. The use of natural composites such as wood, cardboard, hemp fiber composites, bamboo composites, etc. should be examined for effectiveness and appropriateness.

STR 8.2.3. Teams should simulate and/or test the properties of materials and the strength of components developed for use in their satellites. This process should include both individual and joint analyses to ensure the integrity of all parts.

It is important to verify the suitability and durability of all materials and parts under the specific conditions they will face during their missions, as this is essential for ensuring launch safety.

STR 8.2.4. Environmentally hazardous materials shall not be used in the satellite project. This includes lead, mercury, uranium, and any other radioactive materials.

Section 9 — Electronic Power System and Communications (ECS)

Section 9 defines the Electronic Power System and Communications (ECS) guidance and overall requirements for any satellite mission.

9.1. Power Systems and Critical Wiring

Electronic Power Systems and Rail Standby Time

ECS 9.1.1. The on-board power system shall be designed and provided by each team. Teams are not required to design solar panels.

ECS 9.1.2. The on-board power system shall store enough energy to ensure power throughout the test mission or launch campaign.

ECS 9.1.3. In order to overcome critically inefficient use of valuable and limited launch campaign time, including slot availability, satellite projects shall have at least 4 (four) hours of battery endurance.

ECS 9.1.4. Satellites should incorporate battery circuit protection for charging/discharging to avoid unbalanced cell conditions.

Remove Before Flight (RBF)

ECS 9.1.5. Each satellite shall include an RBF pin.

ECS 9.1.6. The RBF pin shall cut all power to the satellite once it is inserted into the satellite.

ECS 9.1.7. The RBF pin shall be removed from the satellite after integration into the experimental launch vehicle.

9.2. Communications and Radio Frequency

ECS 9.2.1. The communication system shall be designed and provided by each team.

ECS 9.2.2. The communication system shall be wireless.

ECS 9.2.3. The satellite's communication system should be capable of establishing reliable communication, and, if necessary, command and control the experimental satellite from an isolated location.

Telemetry, Tracking, and Control (TT&C) systems represent a foundational element in the architecture of next-generation satellites. LASC encourages all participating missions to implement TT&C modules in their experimental satellite designs.

ECS 9.2.4. All wireless communications should use U.S. or Brazilian legal frequencies and shall be used responsibly

Radio Frequency

ECS 9.2.5. Competitors agree that use of Radio Frequencies (RF) for any purpose will be in accordance with U.S. or Brazilian laws and regulations. Each team shall coordinate frequency allocation with the LASC Organizers.

ECS 9.2.6. For all communications, any electromagnetic spectrum frequency (e.g., RF, infrared, visible light, etc.) is allowed, subject to all applicable spectrum allocation.

ECS 9.2.7. Each team is responsible for coordinating RF operating licenses for both their satellite and for all ground stations under their control if required.

Section 10 — Recovery System (REC)

Section 10 defines the Recovery System (REC) guidance for the recovery system, parachute design, and other requirements.

10.1. Guidance

Satellites may remain secured within the payload bay of the experimental rocket and be recovered together with it, or may be independently recovered (e.g., using their own parachute or alternative recovery systems).

REC 10.1.1. The satellite shall be independent of the experimental rocket, with no fixed electrical connection between the two. The satellite shall not command any deployment action of the recovery system or any other type of action.

REC 10.1.2. If a dedicated recovery mission is implemented for the satellite, the satellite team shall plan and define in advance whether the recovery will be conducted using a dependent or independent recovery system.

REC 10.1.3. Satellite missions with a dependent recovery system shall keep the satellite inside the payload bay or airframe throughout the entire launch operation.

REC 10.1.4. If a satellite mission opts for an independent recovery system, the size, mass, and volume of the recovery system shall not be included in the satellite's form factor calculations and measurements.

REC 10.1.5. If a satellite mission decides to use an independent recovery system, the parachute or non-parachute recovery device shall be easily detachable from the structure.

Deployment Event for Satellites with Independent Recovery System

REC 10.1.5. The deployment event for independent payload/satellite recovery shall occur at or lower than 1,500 meters above ground level, with a target descent rate sufficient to permit data collection but not so high as to exacerbate wind drift (e.g., between 20-45 m/s).

REC 10.1.6. The satellite shall be protected to prevent damage from the deployment event initiated by the experimental rocket's recovery electronic system.

10.2. Parachute Design Guidance

It is highly recommended that teams choose traditional parachute-based recovery systems. The majority of flight failures occur during recovery, so keeping it simple is advisable, especially if part of the launch operations score directly or indirectly depends on the successful recovery of the satellite.

Non-Parachute/Parafoil Recovery Systems

REC 10.2.1. Teams exploring other (i.e., non-parachute or parafoil based) recovery methods shall notify LASC of their intentions at the earliest possible opportunity, and keep LASC apprised of the situation as their work progresses. LASC may make additional requests for information and draft unique requirements depending on the team's specific design implementation.

Ejection Gas Protection

REC 10.2.2. The recovery system shall implement adequate protection (e.g., fire resistant material, pistons, baffles) to prevent hot ejection gasses (if implemented) from causing burn damage to retaining cords and other vital components as the specific design demands.

Parachute Coloration And Markings

REC 10.2.3. Parachutes shall be high-contrast to the local environment. It is recommended to use orange, white, and/or red colors. This will enable ground-based observers to more easily characterize deployment events with high-power optics..

Appendix A: Details for the Mission Report

Abstract

The Mission Report shall contain an Abstract. At a minimum, the abstract shall identify the mission category in which the team is competing, identify any unique/defining design characteristics of the project, and provide whatever additional information may be necessary to convey any other high-level project or program goals & objectives.

Introduction

The Mission Report shall contain an Introduction. This section provides an overview of the academic program, stakeholders, team structure, and mission organization and management strategies. The introduction may repeat some of the content included in the abstract, because the abstract is intended to act as a standalone synopsis if necessary.

System Architecture and Concept of Operations

The Mission Report shall contain a System Architecture Overview. This part shall begin with a top-level overview of the integrated system, including a cutaway figure depicting the fully integrated project and its major subsystems or modules – configured for the mission proposed.

The Mission Report shall contain a Mission Concept of Operations (ConOps) Overview. This part shall identify the mission phases, including a figure, and describe the nominal operation of all subsystems or modules during each phase (e.g., a description of what is supposed to be occurring in each phase, and what subsystems are responsible for accomplishing this).

No matter how a team defines mission phases and phase transitions, they will be used to help organize failure modes identified in a Risk Assessment Appendix – described in this document.

Weights, Measures, and Performance Data

The Mission Report shall contain Weights, Measures, and Performance Data. This requirement will be satisfied by informing the data in this section of the document. A table with the minimum data is available in the Mission Report template.

Conclusions and Lessons Learned

The Mission Report shall contain Conclusions and Lessons Learned. This section shall include the lessons learned during the design, manufacture, and testing of the project, both from a team management and technical development perspective. Furthermore, this section should include strategies for corporate knowledge transfer from senior team members to the rising underclassmen who will soon take their place.

Hazard Analysis Appendix

The first Mission Report appendix shall contain a Hazard Analysis Report. This appendix shall address as applicable, hazardous material handling, transportation, storage procedures, and any other aspects of the design which pose potential hazards to operating personnel. A mitigation approach – by process and/or design – shall be defined for each hazard identified.

Risk Assessment Appendix

The second Mission Report appendix shall contain a Risk Assessment. This appendix shall summarize risk and reliability concepts associated with the project. All identified failure modes which pose a risk to mission success shall be recorded in a matrix, organized according to the mission phases identified by the ConOps.

A mitigation approach – by process and/or design – shall be defined for each risk identified. A common description of the Risk Assessment is FMECA (Failure Mode and Effect Criticality Analysis). A Risk Assessment/FMECA is often represented as a spreadsheet matrix. The input to the matrix is listed as follows:

- A description of the identified failure mode.
- The likelihood of the failure mode occurring.
- The severity and impact of the failure mode occurring

The likelihood of a failure mode occurrence and the severity of the occurrence is assigned values according to the following tables:

Table 7: Likelihood of Failure.

Failure Probability	Value	Assessment of Risk
Remote	1	This is unlikely to happen.
Occasional	2	This might happen.
Probable or likely	3	This is likely to happen.

Table 8: Severity of Occurrence.

Mishap Severity	Value	Effect of Failure Mode
Minor or negligible	1	Minor impact on mission.
Critical	2	Deterioration of performance and mission.
Catastrophic	3	Safety hazard and/or likely loss of mission

The "Criticality Ranking" is the product of the Failure Probability and the Mishap Severity. The criticality rating is a measure of how urgent and how severe mitigation actions will have to be taken, to reduce the Criticality Ranking.

Table 9: Criticality Ranking.

Mishap Severity	Value	Effect of Failure Mode
1	Minor	This failure mode is not a concern.
2	Minor	This failure mode is of very minor concern.
3	Medium	Justification needed. Jury may decide to review.
4	High	Technical jury approval needed before launch.
6	Critical	Action required to reduce ranking before launch.
9	Critical	Action required to reduce ranking before launch.

The output of the matrix is highlighting and ranking failure mode liabilities to the mission, and the justifications and mitigations to reduce the Criticality Ranking. An example is given below:

Table 10: Risk Matrix.

Failure Mode	Mission Phase	Failure Probability	Mishap Severity	Critically Ranking	Mandatory Comments
Battery Thermal Runaway	Pre-launch phase	1	3	3	A thermal runaway begins when the heat generated within a battery exceeds the amount of heat that is dissipated to its surroundings. A battery balancing system will be implemented to reduce the risk.
Antenna Deployment failure	Deployment phase	2	3	6	Antenna deployment is mandatory for communication. A redundant system is required.

A typical FMECA scale for the complexity of projects attending LASC should feature no less than 10 identified, ranked, commented, and justified failure modes – these should address at the minimum all important and critical failure modes.

Engineering Drawings and Optional Appendix

The final Mission Report appendix shall contain Engineering Drawings and optional appendices. This appendix shall include any revision controlled technical drawings necessary to define significant subsystems or components, and other optional appendices can include, but are not limited to further Subsystem Details, Detailed Structural and Mechanical Calculation, Detailed Logical Process Diagrams, Detailed Software Architecture, and Detailed Electrical Architecture.

Appendix B: Detailing Grading Criteria

The grading will be conducted by the event officials based on the individual grading criterion in the respective mission categories. A summary and overview of the grading scheme is given below for clarity.

Table 11: Grading Scheme.

Count	For Countable/Relative Criteria	Count	For Absolute Criteria
91 to 100%	Outstanding Quality/Conformity	100%	Yes
76 to 90%	High Quality/Conformity	0%	No
51 to 75%	No Greater than Average		
Up to 50%	Unsatisfactory		

This is meant to be an intuitive and transparent scheme for the jury to follow and the teams to understand.

Team Effort

Team Effort will be graded up to 100 points (10% of 1,000 points possible). The total points for Team Effort is a combination of countable, relative and absolute criteria.

Table 12: Team Effort Grading Criteria.

Criterion	Outstanding	High	Average	Unsatisfactory	Points
1st Progress Update	5 points Compliant to Section 1.4.	-	-	0 point One or more required items missing.	/5
2nd Progress Update	5 points Compliant to Section 1.4.	-	-	0 point One or more required items missing.	/5
3rd Progress Update	5 points Compliant to Section 1.4.	-	-	0 point One or more required items missing.	/5
4th Progress Update	5 points Compliant to Section 1.4.	-	-	0 point One or more required items missing.	/5
Mission Patch	21-25 points Completely complies with guidance. Excellent quality and design.	16-20 points Complies with the guidance with a few design or quality issues.	10-15 points Minimally complies with guidance. Medium quality and design.	<10 point Does not comply with guidance.	/25
Pitch Video	38-45 points Completely complies with guidance. Excellent quality, & clarity,	31-37 points Complies with the guidance with a few minor issues. High video and/or sound quality.	26-30 points Minimally complies with guidance. Medium video and/or sound quality. < 10% over or < 20% under time limit.	< 25 points Does not comply with guidance. Low video and/or sound quality. > 10% over or > 20% under the time limit.	/45

Criterion	Outstanding	High	Average	Unsatisfactory	Points
Lights, Camera, Action!	The Lights, Camera, Action! points will be normalized from the highest achieving team/selected mission to the lowest. The minimum requirement for not being zeroed is participation with at least 10% of the maximum links requested as stated in Section 1.4.				/10
Team Effort					/100

Mission Report

The Mission Report will be graded up to 200 points (20% of 1,000 points possible) divided into three criteria:

- Completeness (20 points), Correctness (40 points) and Analysis (140 points).
- Completeness Criterion will be scored using Absolute Criteria.
- Correctness and Analysis Criteria will be scored in a mix of Countable and Relative Criteria.

Table 13: Mission Report Grading Criteria - Completeness Criterion.

Criterion	Outstanding	High	Average	Unsatisfactory	Points
Completeness	20 points All required items of the Appendix A present.	-	-	0 point One or more required items missing.	/20

Table 14: Mission Report Grading Criteria - Correctness Criterion.

Criterion	Outstanding	High	Average	Unsatisfactory	Points
Style	18-20 points Writing was exceptionally clear, understandable, and concise. Sentence and paragraph organization is exceptional. Writing is free of digressions or irrelevant information.	15-17 points The writing was clear, understandable, and concise. Overall paragraph and sentence organization were very good. Digressions or irrelevant information do not significantly detract from the report	10-14 points Writing was generally clear and understandable. Paragraph and sentence organization were generally good. Digressions or irrelevant information detract from the report's analysis.	< 10 points Writing was repeatedly unclear, difficult to understand or wordy. Overall paragraph and/or sentence organization were ineffective or nonexistent. Digressions and/or irrelevant information consistently detract from the analysis.	/20
Mechanics	10 points No grammar, spelling, or mechanics errors. Scientific terms correctly used, units and dimensions consistent and correct.	8-9 points No more than a few grammar, spelling, or usage errors. Only a few minor errors with use of scientific terms or dimensions.	5-7 points Significant spelling, usage, and grammar errors that did not detract from readability. Significant errors with use of scientific terms or dimensions.	< 5 points Repeated grammar or spelling errors detracted from readability. Errors with use of scientific terms or dimensions detracted from the report.	/10
Format	10 points Completely follows the required template. Meets page limits.	8-9 points Minor deviations from required template. Meets page limits.	5-7 points Major deviations from required template. < 10% over page limits.	< 5 points No attempt at cohesive format or use of required template. More than 10% over page limits.	/10
Correctness					/40

Table 15: Mission Report Grading Criteria - Analysis Criterion.

Criterion	Outstanding	High	Average	Unsatisfactory	Points
Depth of Analysis	46-50 points Very complete and thorough analysis. All key design decisions are discussed and based on design targets, constraints, and appropriate tradeoffs.	38-45 points Adequate analysis with minor weaknesses. Most key design decisions are discussed and based on design targets, constraints, and appropriate tradeoffs.	25-37 points Adequate analysis with significant gaps or weaknesses. Some key design decisions are discussed and based on design targets, constraints, and appropriate tradeoffs. Some minor incorrect statements.	< 25 points Inadequate analysis. Few, if any key design decisions were discussed. No discussion of tradeoffs. Parts of analysis conflict with general scientific knowledge.	/50
Assumptions and Sensitivity Analysis	27-30 points All assumptions are clearly stated. Sensitivity analysis is performed to quantify uncertainty in variables and assumptions.	23-26 points Most assumptions were addressed. Some sensitivity analysis.	15-22 points Unstated assumptions. No sensitivity analysis.	< 15 points No stated assumptions or assumptions were unreasonable. No sensitivity analysis.	/30
Verification and Validation Tests	36-40 points All verification and validation tests were discussed, both for the final design and key iterations leading to that design. Complete and valid conclusions were drawn from the results.	30-35 points Most verification and validation tests are adequately discussed. Appropriate conclusions were drawn from the results, but key iterations prior to final design were not discussed.	20-29 points Some verification and validation tests are discussed but consistent. Unclear that conclusions and decisions were drawn from testing results and analysis.	< 20 points Unclear whether verification and validation tests were performed. Decisions and conclusions were not drawn from the analysis.	/40
Use of Charts and Figures	18-20 points Tables, figures, and appendices all effectively organize and communicate information.	15-17 points Use of tables, figures, and appendices is mostly effective.	10-14 points Use of tables, figures, and appendices is somewhat effective with significant issues.	< 10 points Tables, figures, and appendices were incorrect or misleading.	/20
Analysis					/140

Then, the *total score* of the Mission Report will be the sum of the points for Completeness, Correctness and Analysis.

Design Implementation

The Design Implementation will be graded up to 100 points (10% of 1,000 points possible) divided into two criteria: Design Quality & Decisions (50 points) and Build Quality 50 points). Both will be scored in a mix of Countable and Relative Criteria.

Table 16: Design Implementation Grading Criteria - Design Quality & Decisions Criterion.

Criterion	Outstanding	High	Average	Unsatisfactory	Points
Team Design Vision, Goals and System	26-30 points A clearly understood design vision with	20-25 points Design vision is mostly understood and	15-19 points Design vision is incomplete or questionably achievable.	< 15 points Questionable or unachievable design	/20

Criterion	Outstanding	High	Average	Unsatisfactory	Points
Engineering	achievable goals, aligned with strategic objectives, and strong evidence of systems engineering discipline throughout the design team.	achievable with coherent goals. Key project elements address team goals. Good systems engineering discipline throughout development. Most of the design team supports coherent and understood goals.	Unclear how project elements address team goals. Some lapses in systems engineering discipline. Inconsistent support for team goals and some evidence of design team working at cross-purposes.	vision. Most project elements do not address team goals. Major lapses in systems engineering discipline. No unified design goals, with clear evidence of team members working at cross-purposes.	
Team Knowledge	9-10 points Strong team understanding of the principles governing design and reasoning behind the design. All members of the team can clearly articulate reasoning for choices.	7-8 points Generally good team understanding of the physical principles governing design and reasoning behind the design. Team members defer to a few team "experts" during discussion.	6-5 points Some team understanding of the physical principles governing design and reasoning behind the design. Team members defer to one or two team "experts" during discussion.	< 5 points Inadequate team understanding of the principles governing design and reasoning behind the design.	/10
Presentation Session	20 points Completely complies with Section 1.4.	10 points Complies with Section 1.4. with minor issues.	5 points Minimally complies with Section 1.4.	0 point Does not comply with Section 1.4.	/20
Design Quality & Decisions					/50

Table 17: Design Implementation Grading Criteria - Build Quality Criterion.

Criterion	Outstanding	High	Average	Unsatisfactory	Points
Compliance with LSARPs	13-15 points Completely complies with LSARPs	10-12 points Complies with LSARPs with a few minor issues.	7-9 points Minimally complies with LSARPs.	< 7 points Does not comply with LSARPs.	/15
Design Quality and Robustness	13-15 points Design and build quality are robust and sufficient to operate as intended under reasonably expected conditions.	10-12 points Design and build quality are somewhat robust and sufficient to operate as intended under reasonably expected conditions.	7-9 points Design and build quality are sufficient to operate as intended under specific conditions but are not robust to reasonably expected variations.	< 7 points Design and build quality insufficient to operate as intended under expected conditions. No attempts at robust design.	/15
Fabrication and Construction Methods	9-10 points Fabrication and assembly methods are fully understood and correctly applied. SRAD manufacturing methods are appropriate and well understood, including cost, time, and performance.	7-8 points Fabrication and assembly methods are generally well understood and correctly applied. Manufacturing methods for SRAD elements are both appropriate and reasonably understood by the team, including cost, time, and performance.	5-6 points Fabrication and assembly methods are appropriate, but not completely understood. Manufacturing methods for SRAD elements are appropriate, but not fully understood by the team.	< 5 points Fabrication and assembly methods inappropriate or not understood. Manufacturing methods for SRAD elements are impractical or not well understood by the team.	/10
Consistent Design (30 pts)	9-10 points Clearly consistent with the team's vision. No evidence of key	7-8 points Generally aligned with the team's vision. No evidence of key systems added as an	5-6 points Somewhat aligned with the team's vision. Some key systems added as afterthoughts.	< 5 points No apparent organizing vision. Key systems added as field modifications or	/10

Criterion	Outstanding	High	Average	Unsatisfactory	Points
	systems added as an afterthought.	afterthought.		afterthoughts.	
Build Quality					/50

Technology and Operations Readiness Level

Each mission will be awarded as many as 550 points – 55% of 1,000 points possible – for their presentation and satellite launch operations at the Latin American Space Challenge, demonstrated by technology and operation readiness, successful communication and recovery.

The Technology Readiness constitutes 250 points of the total score assigned to the Technology and Operations Readiness Level. This includes completing an oral presentation, worth 100 points, and the successful demonstration of the minimum required objectives for a satellite during a workbench test, also worth 100 points (i.e., turning the satellite power on and demonstrating a real-time data collection via wireless or wired communication).

The Operation Readiness constitutes 50 points of the total score assigned to the Technology and Operations Readiness Level. This includes completing the Launch Readiness Review (LRR) with a green/approval flight status, worth 25 points, and the successful integration with an experimental rocket, also worth 25 points (i.e., completing the integration of the satellite with the experimental rocket).

The Successful Communication constitutes 150 points of the total score assigned to the Technology and Operations Readiness Level. This includes completing at least one receiving packet of data from the satellite during the launch operations, worth 50 points, and the successful data collecting from the satellite during the flight, also worth 50 points (i.e., collect data from the satellite during boost, coasting, apogee and descent phase). An additional 50 points will be assigned to satellite missions that demonstrate a wireless half or full-duplex communication.

The Successful Recovery of the satellite is worth 100 points of the overall value assigned to launch operations. A recovery operation is considered successful according to the following table of classification:

Table 18: Launch Operations Status and Recovery Points.

Criterion	Outstanding	Recovery Points
No Launch	A Selected Mission either did not request a Launch Readiness Review (LRR), received a Launch Status equals to "orange" or "yellow" during the event, or received a "green" but did not attempt at least one ignition.	Zero
Launch Denied	A Selected Mission did not pass the Launch Readiness Review (LRR) due to one or multiple criteria being indicated as "red."	Zero
Operational or Launch Failure	The satellite suffered an indirect failure (e.g., Catastrophe At Take Off, No Ignition/Ignition Failure, etc.) or has not been found after an abnormal trajectory during the available recovery slots during the event.	Zero
Not Recovered	The satellite has not been found after the nominal trajectory during the available recovery slots during the event.	Zero

Criterion	Outstanding	Recovery Points
Extensive Damage	Excessive damage is defined as any damage to the point that, if the systems intended consumables were replenished, it could not be launched again safely.	Zero
None or Minor Damage	The satellite has been successfully recovered after a nominal flight with no or minor damages (i.e., if the system's consumables were replenished, it could be launched again safely).	100

Dual-Challenge Bonus

A sounding rocket or a launch vehicle transports payloads for research and development, commercial purposes, education or training. An experimental rocket without a payload loses its purpose. The Latin American Space Challenge promotes the development of experimental satellites as part of the mission's payload.

Teams with Selected Missions in both the Rocket and Satellite Challenge *may* receive a bonus of 25 points if, and only if, the experimental satellite presented in the Satellite Challenge is placed in the experimental rocket and a launch is performed (i.e., the ignition of the propulsion system is initiated).

If two different teams coordinate to fly a satellite in the same experimental rocket, the Dual-Challenge Bonus may also be granted. However, both teams must inform event officials of this intent during the Launch Readiness Review (LRR)

Final Score

The maximum points possible are up to 1,000 points, including the bonus points of participating with Selected Mission in both Rocket and Satellite Challenge.

$$\text{Final Score} = (\text{Team Effort} + \text{Mission Report} + \text{Design Implementation} + \text{Technology and Operations Readiness Level}) + \text{Bonus Points} - \text{Penalties}$$

